WEEK 1 · DAY 5 · OVERVIEW

Case Study

A New Issue Calendar



Created by **Jeff Lenamon**

ABOUT 40 MIN WITH EXERCISES / 6 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Definition

Eight new issues price today, all settling on 30 September 2026. Before the calendar goes round: what does each yield at its re-offer price? Price all eight at one yield, so coupon and term compare on equal terms. On the grid of eighths, which coupon would price each one nearest par at its own yield, and what is one eighth of coupon worth in price? Last, send it as a workbook whose yields and prices Excel works out itself. Describe the calendar: no view on any issue.

Your Route Through Day 5

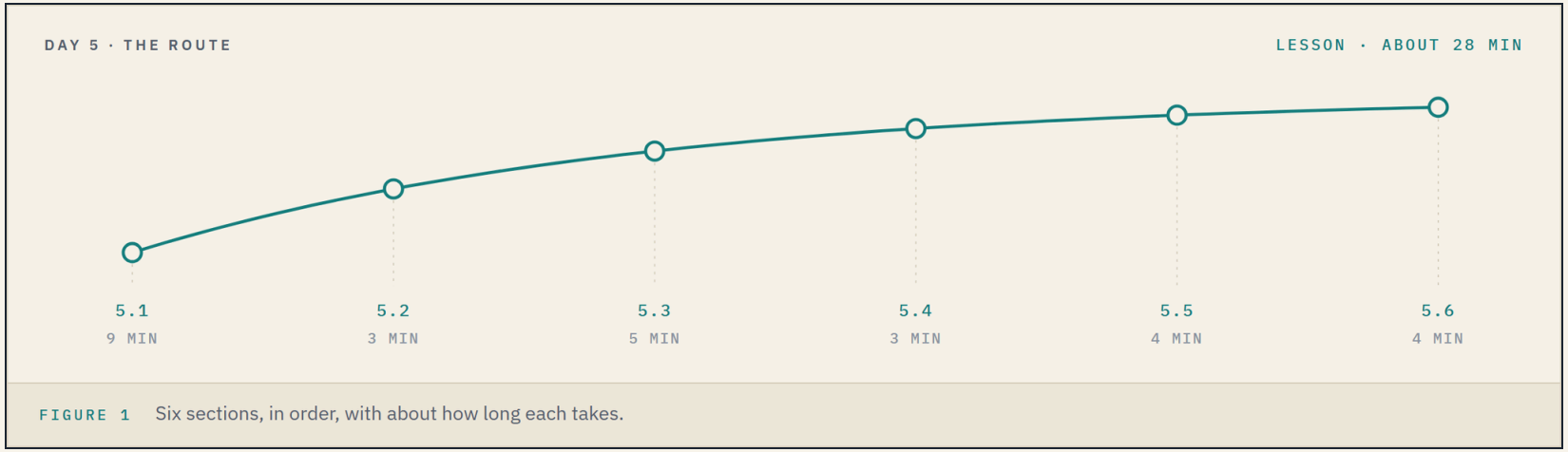


FIGURE 1 Six sections, in order, with about how long each takes.

	SECTION	ABOUT
5.1	Yields From Re-offer Prices	9 min
5.2	Prices at a Common Yield	3 min
5.3	The Coupon That Prices Nearest Par on a 1/8 Grid	5 min

	SECTION	ABOUT
5.4	The Price Effect of a 1/8 Coupon Step	3 min
5.5	Two Rules Into the Test File, and the Week Tagged	4 min
5.6	The Workbook for Excel	4 min

Time. About 28 minutes for the lesson, then about 12 for the four exercises.

5.1 Yields From Re-offer Prices

YIELD OF ONE ISSUE

```
thalostrin = calendar.iloc[0]
thalostrin_yield_pct = bondmath.yield_from_price(thalostrin['coupon_pct'],
                                                 thalostrin['reoffer_price'], thalostrin['years'])
print(f"Thalostrin yields {thalostrin_yield_pct:.4f} percent at {thalostrin['reoffer_price']}")
```

```
Thalostrin yields 5.1539 percent at 99.874
```

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
Thalostrin's yield	`=YIELD(DATE(2026,9,30),DATE(2031,9,30),5.125/100,99.874,100,2,0)`	0.0515390782041645

Eight yields, each within 6.1e-09 percent of Excel. Now take credit out: one yield for all.

5.2 Prices at a Common Yield

THE COMMON YIELD

Price all eight at one round yield, 5.50 percent, a what-if and not a market level, and only coupon and term are left.

PRICE ALL EIGHT AND COMPARE WITH EXCEL

```
calendar['price_at_5_50'] = [bondmath.price_from_yield(c, common_yield_pct, t)
                             for c, t in zip(calendar['coupon_pct'], calendar['years'])]
excel_prices = tvm[tvm['function'] == 'CALENDAR_PRICE'].head(8)
calendar['excel_price_at_5_50'] = excel_prices['excel_value'].to_numpy()
largest_gap = (calendar['price_at_5_50'] - calendar['excel_price_at_5_50']).abs().max()
print(f'Largest gap to Excel: {largest_gap:.1e} per 100 of face')
```

```
Largest gap to Excel: 1.1e-13 per 100 of face
```

A coupon above 5.50 then prices over 100, one below it under 100 (notebook 03).

5.3 The Coupon That Prices Nearest Par on a 1/8 Grid

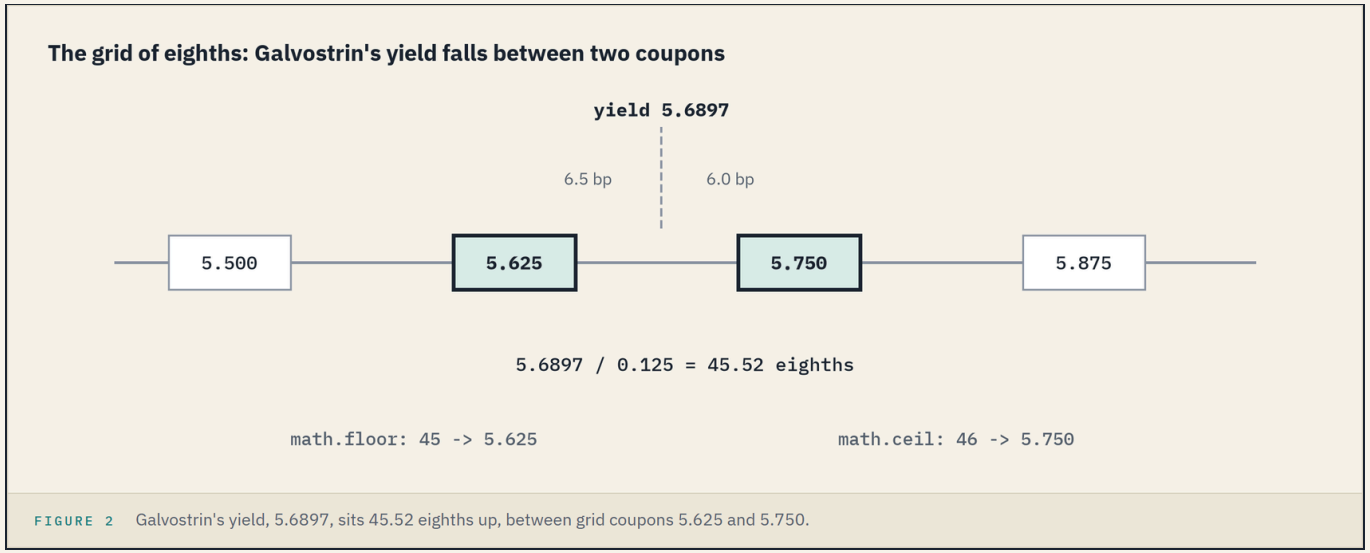


FIGURE 2 Galvostrin's yield, 5.6897, sits 45.52 eighths up, between grid coupons 5.625 and 5.750.

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
Round down to the grid	`=FLOOR(5.69,0.125)`	5.625
Round up to the grid	`=CEILING(5.69,0.125)`	5.75

ROUND BOTH WAYS

```
low_coupon, high_coupon = math.floor(eighths) * 0.125, math.ceil(eighths) * 0.125
```

```
5.6897 percent is 45.52 eighths: 5.625 or 5.75
```

So take the two grid coupons either side of the yield, and keep the one that prices nearer 100.

5.4 The Price Effect of a 1/8 Coupon Step

THE SAME BOND PRICED TWICE

```
price_now = bondmath.price_from_yield(6.0, common_yield_pct, 30)
price_stepped = bondmath.price_from_yield(6.125, common_yield_pct, 30)
```

AN EIGHTH ON EVERY ISSUE

```
calendar['step_value'] = [bondmath.price_from_yield(c + 0.125, common_yield_pct, t)
                          - bondmath.price_from_yield(c, common_yield_pct, t)
                          for c, t in zip(calendar['coupon_pct'], calendar['years'])]
excel_stepped = tvm[tvm['function'] == 'CALENDAR_PRICE'].tail(8)['excel_value'].to_numpy()
excel_steps = excel_stepped - calendar['excel_price_at_5_50']
print(f"Largest gap to Excel: {(calendar['step_value'] - excel_steps).abs().max():.1e}")
```

```
Largest gap to Excel: 2.8e-14
```

Thalostrin and Corvolvey, both 5 years, share one dot although their coupons are almost two points apart: on a straight line the step does not depend on the coupon.

5.5 Two Rules Into the Test File, and the Week Tagged

A tag never moves; a branch moves on with each commit

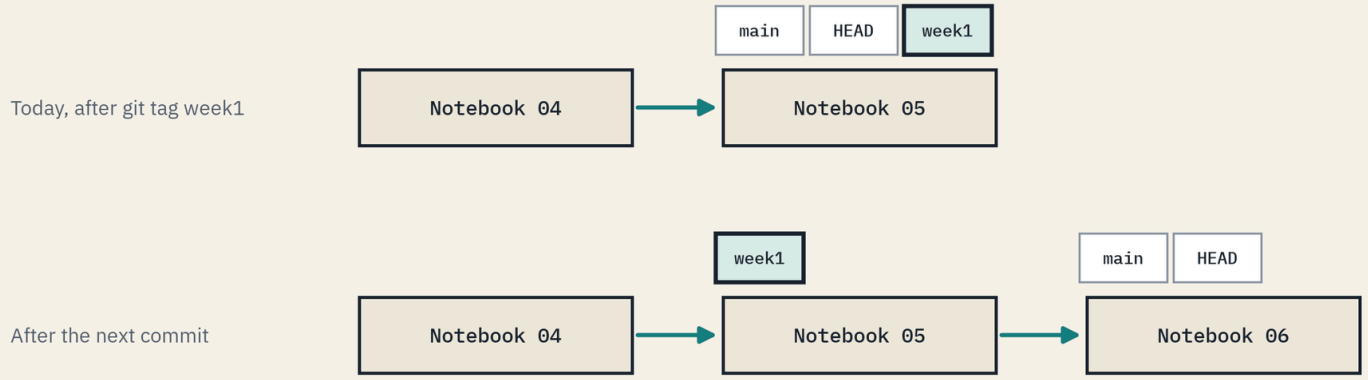


FIGURE 4 Tag week1 stays on Notebook 05 while main and HEAD move on to Notebook 06.

WHAT GIT SEES

```
!git -C "{work}" status --short
!git -C "{work}" diff --stat

M test_bondmath.py
?? __pycache__/
 test_bondmath.py | 15 +++++++++++++++
 1 file changed, 15 insertions(+)
```

TAG THE WEEK

```
!git -C "{work}" tag week1
!git -C "{work}" log --oneline --decorate
```

13 tests pass and `week1` marks the module as the week ends. Last ask: the workbook.

5.6 The Workbook for Excel

FORMULAS INTO A WORKBOOK

```
sheet['excel_yield_pct'] = [f'=YIELD(DATE(2026,9,30),D{r},C{r}/100,E{r},100,2,0)*100'
                            for r in rows]
sheet['yield_gap'] = [f'=G{r}-F{r}' for r in rows]
                                for r in rows]
sheet['price_gap'] = [f'=J{r}-I{r}' for r in rows]
print(sheet.loc[0, 'excel_yield_pct'])
```

```
=YIELD(DATE(2026,9,30),D2,C2/100,E2,100,2,0)*100
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL	RESULT
Yield, row 2	`yield_pct`	`=YIELD(DATE(2026,9,30),D2,C2/100,E2,100,2,0)*100`	5.153908
Price at 5.50, row 2	`price_at_5_50`	`=PRICE(DATE(2026,9,30),D2,C2/100,5.5/100,100,2,0)`	98.379986

Excel calculated the file itself and agreed to 6.1e-09 percent. The note is answered.

You Can Now

- Turn a calendar of re-offer prices into yields, one bond at a time.
- Price a set of bonds at one common yield, and say why a long small premium can outweigh a short large one.
- Find the coupon on the grid of eighths that prices nearest par.
- Measure what an eighth of coupon is worth, and say why it is under an eighth a year.
- Write a workbook of live **`YIELD`** and **`PRICE`** formulas, and say why pandas reads them back empty.
- Name a commit with **`git tag`**, and read the names with **`git log --oneline --decorate`**.

NEXT

WEEK 2 · DAY 1

Dates and Coupon Schedules

Bonds now settle between coupons: dates in Python, moving a date by months, and a coupon schedule counted back from maturity, tied to Excel's COUPPCD, COUPNCD, and COUPNUM.

